Sourcegraph

Sourcegraph MCP Best Practices

Sourcegraph MCP

The Sourcegraph Model Context Protocol (MCP) Server provides AI agents and applications with programmatic access to your Sourcegraph instance's code search, navigation, and analysis capabilities through a standardized interface.

Supported Clients

- Amp
- Claude Code
- OpenCode
- VsCode (CoPilot)
  - Cursor
  - Antigravity

Full Documentation: sourcegraph.com/docs

Agents Without & With Sourcegraph MCP

Without

Agent relies only on:

- Files currently in context
- What the user provides in the prompt
- The model's general knowledge

User interaction:

"Explain this function."

"Refactor this file."

"Write a helper for this code."

With

Agent can actively explore the codebase.

- Search across repositories
- Find symbol definitions and references
- Read files across projects
- Investigate system relationships

User interaction:

"Use Sourcegraph MCP to find where this function is used."

"Investigate how authentication works across the codebase."

When to Use Sourcegraph MCP

Use MCP when the agent needs real information from across your codebase.

Good use cases:

- Searching across many repositories
- Reading specific files from remote repositories
- Understanding unfamiliar systems
- Investigating how components interact between services/repositories
- Looking at commit history or changes
- Researching complex questions across an entire codebase

Best Practices

1. Get familiar with the available tools

Before prompting an agent, take time to understand the tools available through the Sourcegraph MCP.

Why this matters:

- You'll know what the agent is capable of doing with the additional tooling
- You can prompt the agent to use the right tool
- You'll get more reliable and faster results

Tools available include:

- Keyword Search
- Natural Language Search
- Go to Definition
- Diff Search
- Commit Search

You can review available tools at the Sourcegraph Docs site.

2. Prompt the agent to use tools

Agents do not always automatically use available tools.

Explicitly encourage the agent to use its MCP tools when investigating a question.

You can do this by mentioning a specific tool, or by simply telling the agent to use Sourcegraph MCP.

Example prompts:

- "Use the Sourcegraph MCP to investigate this issue."
- "When was the indexing queue functionality last changed in @zoekt? Search the last 5 commit diffs touching this code and explain the changes."

Pro Tip:
If the agent answers without using tools, ask it to re-investigate using MCP tools.

3. Ask the agent to gather info first

Encourage the agent to the MCP to ***investigate first,*** before providing an answer.

When given Sourcegraph MCP, the agent becomes an investigator with tools that allow it to perform real research on your behalf.

By prompting the agent to investigate first, it eliminates premature conclusions and limits

Example prompts:

- “Find the services that depend on this component.”
- “Find related files & references for this function and summarize their connection.”
- “Gather relevant information about the authentication services before answering.”

4. Be specific about the system data you want

Provide clear targets when prompting so the agent can choose the right tool and retrieve useful information. You can do this by explicitly referencing specific data you're interested in.

Examples of specific data:

- Repository or service name
- Function or Symbol
- File or directory
- Commit history or changes
- References or dependencies

Example prompts:

- Search the last 5 commit in the @***zoekt repo*** involving the queuing functionality and explain the changes."
- Investigate how the ***payment service*** interacts with the ***user service***."

5. Break complex requests into steps first

AI agents perform best when tasks are broken into clear stages rather than being asked to solve everything at once.

Why it helps:

- Improves accuracy of tool usage
- Makes results easier to verify
- Reduces confusing by overloading the context window

How to do it:

- Ask the agent to gather information first
- Then analyze or summarize what it found
- Finally, produce the answer or recommendation

Key Idea:

Small steps > Better Reasoning > Better Results

6. Ask for evidence from the codebase

When an agent provides an answer, ask it to ***show the codebase evidence that supports it.***

- File paths
- Code snippets
- Symbol references
- Search results

Your agent can leverage Sourcegraph tooling to help build trust that it's solutions are grounded in real code.

Example prompts:

- "Provide the code references that support your answer."
- "Include relevant code snippets from the codebase that supports this change."